

School of Engineering

The University of Warwick

General Assignment Information

Module Code	ES96T
Module Title	ADVANCED WIRELESS SYSTEMS AND NETWORKS MODULE
Assignment Title	Assignment 2025
Assignment Weighting / Credits	40% of the module/6 credits
Submission Deadline	Noon, Tuesday, Week 32

Intended Learning Outcomes (ILOs) Assessed

Intended Learning Outcome(s) (ILOs)	Task / Criterion / Section
Design efficient networks in terms of energy and/or cost using communication network modeling and optimisation theory	Task 1
Using simulation software (i) design cellular networks and analyse their performance through modelling their mutual interference interactions, (ii) optimise network design in wireless local area and mobile ad-hoc networks for efficient congestion control and reliable data transfer.	Task 2 & 3

Notes:

- A total mark below 40% indicates that the ILOs have not all been met at threshold level;
- A total mark in the range 40 – 48% indicates that the ILOs have all been partially met to at least threshold level;
- A total mark of at least 50% indicates that the ILOs have all been met.

Submission Details

The submission details for this assignment are:

- **Deadline:** Noon, Tuesday, Week 32
- **Method:** Online submission via Tabula.
- **Format of submission:** Electronic reports along with MATLAB/Python scripts must be included in a common ZIP folder. A meaningful filename that would contain the student number, the module code, and assignment name, for example: "1234567_ES96T_Lab Report.zip" is recommended.
- **Submission length:** 8 pages for main report. Extra pages could be used for appendices if required (e.g., to submit additional plots, etc.).
- **Formatting instructions:** Use a minimum 11-point Arial (or equivalent) font for the text, with 1.5 line spacing and 25 mm margins all round.

Note: Submissions should be of an appropriate file size and students are responsible for ensuring that work is uploaded successfully before the deadline. If there are technical issues when submitting online, please contact the Engineering Student Office (eng.eso@warwick.ac.uk).

Guidance and Referencing Style

It is serious Academic Misconduct to pass off the work of others (including peers or AI-based chatbots such as ChatGPT) as your own and you should not permit colleagues to copy from you. Sources must be appropriately and properly acknowledged every time reference is made to another's work, using the Harvard Referencing system. Failure to do so amounts to plagiarism which breaches university regulations and falls short of the Academic Integrity expected in the department and university.

Find out more about the School of Engineering Referencing System here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/sb-2-04

There are also other types of academic offences including duplication or 'self-plagiarism'. Refer to <https://warwick.ac.uk/fac/arts/history/students/undergraduate/assess-plagiarism/> for further details.

Style and Formatting Guide

Submissions are expected to conform to professional standards on style and formatting, and guidance can be found here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/sb-1-06/

Assignment Feedback

Students will receive individual feedback on their submission. Feedback on students' assignments will be provided through the Tabula system, together with the release of their awarded marks. The feedback will be comprehensive, consisting of an annotated report which highlights the strengths of the submission, identifies the areas for improvement, and provides comments for students to enhance their future work.

The Brief

The assignment will contain a mix of numerical and essay-type questions, and MATLAB / Python programming exercises related to the lecture contents. The questions are expected to test the understanding of the course material as they require application of the underlying concepts. **Each student must work individually and are expected to submit their own report and programming scripts.**

The assignment will consist of three tasks one for each Topic (the headings of which are given below). Task 1 will be the test student's understanding of *Radio Resource Allocation in 4G cellular networks*. Task 2 will test the student's understanding of *performance analysis of CSMA/CA mechanism in IEEE 802.11 communication systems*. Finally, Task 3 will be an essay-type question will test the student's ability to understand and summarize the latest research trends in wireless communications field.

Marking Scheme:

Task 1 (25 marks) and Task 2 (45 marks) and Task 3 (30 marks): Tasks 1-3 are numerical questions along with MATLAB exercises, which will be broken down into sub questions with small marks for each question, and marks will be allocated accordingly based on technical accuracy/completeness/clarity.

Topic 1: Channel Fading in Wireless Communications (25 marks)

Task 1.1: A cellular communication system follows the “Hexagonal cell layout” pathloss model. The carrier wave frequency is 1 GHz and the base station transmits at 1 W. What is the pathloss in dB scale at a distance of 100 m from the base station?

(5 marks)

Task 1.2: If the minimum acceptable signal level for coverage is given by -104.1 dB, compute the coverage probability, P_c , and outage probability (i.e., the user not being in coverage) P_{out} at a distance of 100 m from the base station?

(8 marks)

Task 1.3: Write a MATLAB script that takes the carrier wave frequency, base-station transmit power, distance between the base station and mobile station, and the standard deviation of the shadow fading component as inputs and implements the Hexagonal cell layout model of Question 1(a).

(6 marks)

Task 1.4: Further, implement 1000 different realizations of the shadow fading component and compute the received signal power for each shadow fading realization for a mobile station located 100 m from the base station. What is the probability of coverage computed from your simulations? Match this with the theoretical value derived in Question 1(b).

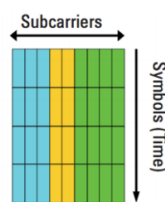
(6 marks)

Topic 2: Radio Resource Allocation in 4G cellular networks (45 marks)

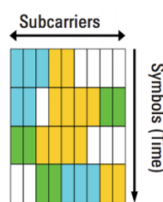
4G Cellular networks are based on orthogonal frequency division multiple access (OFDMA), in which the available wideband frequency resource is broken down into multiple smaller sub-bands with the flexibility of allocating it to different users (as shown in the slide below).

4G Multiple Access and MCS: OFDMA

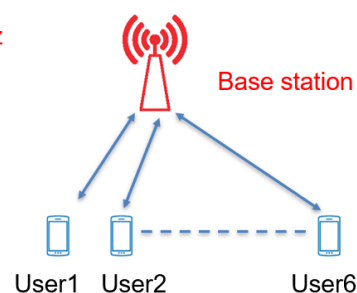
- Division of frequency resource



1/3/2024 OFDM



OFDMA



7

The optimal radio resource allocation (i.e., assigning subchannels to the users and allocating power over them) is setup as an optimisation problem. The basis for solving this problem is setup as multi-user water filling problem, which consists of base station power allocation across channels of varying strength cast as the following optimisation problem [1].

The contents of this power allocation are drawn from recent research articles published in IEEE conferences as shown below. The papers are uploaded on Moodle and available for students to read them in detail (they will be covered in the class).

Multiuser Water-filling in the Presence of Crosstalk

Wei Yu

Department of Electrical and Computer Engineering
University of Toronto
Toronto, Ontario M5S 3G4, Canada
Email: weiyu@comm.utoronto.ca

Abstract—Spectrum optimization is an important part of the design of interference-limited multiuser communication systems. While traditional water-filling provides a closed-form solution to the transmit optimization problem in a single-user system, the multiuser version of the problem often leads to nonconvex problem formulations which are difficult to solve. Motivated by high-speed parallel-link and digital subscriber line applications, this paper investigates two practical multiuser settings in which global or local optimal solutions to the multiuser spectrum optimization problems can be found efficiently. The first part of this paper considers a high-speed transmission system in which practical (but suboptimal) minimum-mean-squared-error linear equalizers (MMSE-LEs) are used at the receiver. The optimal single-user transmit spectrum in this case involves a modified water-filling solution. Surprisingly, such a modified water-filling spectrum can be shown to be near-optimal in a multiuser setting as well, if the direct-link and the crosstalk characteristics are symmetric and if crosstalk is reasonably small. Thus, for practical parallel-link systems using MMSE linear equalizers, the optimal single-user and multiuser spectra are nearly identical. The second part of this paper considers numerical techniques for solving a nonconvex multiuser rate maximization problem for digital subscriber line applications. A new ingredient in the proposed approach is a taxation scheme that takes into account the effect of interference between neighboring lines. This leads to a modified iterative water-filling algorithm which is capable of finding local optimum solutions to the multiuser spectrum optimization problem efficiently.

multiuser setting as well. In the second half of this paper, we focus on efficient numerical solutions to a nonconvex multiuser rate maximization problem in the presence of crosstalk interference. We show that a modified iterative water-filling algorithm can be used to arrive at local optimal solutions of the problem efficiently.

We are motivated to study the multiuser transmit spectrum optimization problem by high-speed parallel transmission line and digital subscriber line applications. (See Fig. 1.) In high-speed backplane communication systems, data transmissions occur on parallel links. The crosstalk between the links is a major transmission impairment. In these high-speed applications, the information-theoretical optimal multi-carrier modulation or decision-feedback equalization may be too complex to implement; suboptimal linear equalizers are often used. The first part of this paper tackles the optimal transmit power allocation problem in the presence of crosstalk when MMSE linear equalizers are used at the receiver.

The second part of the paper is motivated by applications in which information-theoretic achievable rate is the correct metric to optimize. This is the case for digital subscriber line systems where optimal receivers are used. The multiuser rate maximization problem for an interference channel is a well-known nonconvex optimization problem. This paper provides

Task 2.1 Construct a 2-dimensional simulation space consisting of **two cells**, each of dimension 1 km x 1 km. Place a base station at the center of the cell. Then, place a single user terminal per cell whose location is chosen according to a uniform random distribution within the cell. Plot the site locations in a plot with different markers (e.g., triangle for BSs, and circle for UTs).

(5 marks)

Task 2.2 Assume that the BSs of all the cells transmit at a power of 10 dBW and the noise power is -114 dBm. The BS divides the available bandwidth into 5 subchannels. Consider the channel path loss model to be given by $PL = 36.7 \log_{10}(d) + 22.7 + 26 \log_{10}(fc)$ and a carrier frequency of 2.4 GHz. The channel model must also account for slow fading (shadowing) which is assumed to be independent across the subchannels with $\sigma = 4$. The BS divides the power between the subchannel according to the iterative water filling algorithm. Write a MATLAB code to do the following.

(a) Implement the proposed channel model.

(5 marks)

(b) Implement the iterative multi-user water filling based power allocation algorithm. You must follow the expressions of Reference [1].

(25 marks: Efficient Code - 20 marks + Detailed comments - 5 marks)

Task 2.3 Using the results generated from Task 1.2, write a MATLAB script to generate the following:

(a) Compute the sum rate that each user derives.

(5 marks)

(b) A plot of the power allocation as a function of the channel index.

(5 marks)

References:

[1] W. Yu, "Multiuser water-filling in the presence of crosstalk," in Proc. Information Theory and Applications Workshop, Jan 2007, pp. 414–420.

Topic 3: Performance Evaluation of CSMA/CA in IEEE 802.11 Wireless Networks.

This coursework focuses on the performance evaluation of the CSMA/CA medium access protocol used in IEEE 802.11 wireless local area networks. You are expected to consolidate your understanding of how CSMA/CA enables multiple wireless stations to share a common radio channel efficiently.

Before starting the assignment, you should carefully review the relevant lecture slides and recorded lectures on WLAN channel access mechanisms, with particular emphasis on backoff procedures, collision handling, and throughput behavior.

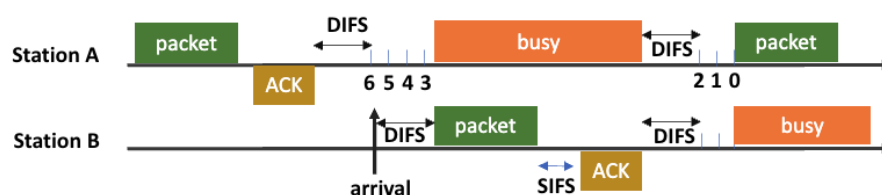


Figure: Basic mode of CSMA/CA mechanism (see details in lecture slides)

In this assignment, you will perform a numerical performance analysis of IEEE 802.11 using a simplified analytical model. You will also demonstrate skills of using basic machine learning method to predict the throughput. A Matlab template will be provided to you. The template implements key elements of Bianchi's analytical model, which allows estimation of network throughput as a function of the number of contending stations.

The following resources are provided via Moodle:

1. Bianchi's original paper on IEEE 802.11 performance analysis

2. A partially completed Matlab implementation
3. The IEEE 802.11a-1999 standard

You need to submit:

1. A report including responses to all tasks.
2. Completed Matlab code for Task 1 (file name format: YOUR NAME_Task1.m)
3. Dataset for Task 3.1 (file name format: YOUR NAME_Task3_1.csv)
4. Python code Task 3.1 (file name format: YOUR NAME_Task3_2.py)

Task 3.1: CSMA/CA Performance Analysis in Basic Access Mode (12 marks)

In this task, you will finalize the Matlab implementation for the basic CSMA/CA access mechanism and evaluate system throughput.

Task 3.1.1. Initialization of Physical Layer Parameters (2 marks)

Using the IEEE 802.11a standard, complete the initialization of PHY-related parameters in the Matlab script **Bianchi11a.m**.

In your report:

- Describe each parameter you have initialized
- Clearly indicate the relevant tables or sections of the IEEE 802.11a standard from which these values are obtained

```

%***** (1) Your input: initialize 802.11a parameters*****
%OFDM symbol duration
SYMBOL =
% PHY preamble duration in seconds
PHY_PREAM =
% service field length in bits
SERVICE =
% tail length in bits
TAIL =
%***** (1) End of your input*****

```

Task 3.1.2. Transmission Duration for a Successful Access. (2 marks)

Complete the code as below.

```

%***** (2) Your Input: Calculate time slot for successful transmission *****
T_S =
%***** (2) End of your input*****

```

In the report, explain your code. Clearly referring to which equation(s) you have used in Bianchi's paper.

Task 3.1.3. Calculate transmission time for collided channel access. (2 marks)

Complete the code as below.

```

%***** (3) Your Input: Calculate time slot for collided transmission *****
T_C =
%***** (3) End of Your Input *****

```

In the report, explain your code. Clearly referring to which equation(s) you have used in Bianchi's paper.

Task 3.1.4. Calculate the effective throughput. (6 marks)

Complete the code as below.

```
***** (4) Your Input: calculate collision probability, p *****
% Hint: use function fsolve() ***

***** (4) End of Your Input *****

***** (5) Your Input: calculate throughput*****
% Hint: you need to assign a value to Bianchi

***** (5) End of Your Input *****
```

In addition, calculate the throughput estimated by Bianchi's model by changing the values of m and CW_{min} according to Table I. Plot your results in a single graph.

Table I: values for m and CW_{min} .

m	CW_{min}
3	32
5	32
3	128

In the report, explain your code. Clearly referring to which equation(s) you have used in Bianchi's paper.

Note: Please ensure you have appropriate comments for your new code. Run your Matlab code and results will be generated in bianchi_11a.txt. By default, the throughput is given with variable number of stations up to 50.

Task 3.2: Impact of m and CW_{min} (8 marks)

Changing the values of variable m and CW_{min} as in Table I. Analyze your results in detail in the report. Provide as many observations/key conclusions as you can. Include high-quality references to justify your conclusions where possible.

Task 3.3 Develop a Machine Learning Model for Throughput Prediction (10 marks)

In this work, Bianchi's analytical model is used to generate synthetic data for training and validating machine learning-based throughput prediction models.

Task 3.3.1 Synthetic Dataset Generation (4 marks)

Using your completed Matlab implementation from Tasks 3.1 and 3.2, generate a dataset suitable for machine learning. You should:

- Vary parameters:

- Number of stations
- $CW_{\min}$
- m
- Data rate (e.g., selected IEEE 802.11a modes)
- Payload and slot duration (e.g. SIFS, DIFS, packet size.) – optional.
- For each parameter combination, calculate and record the throughput. Note that the size of the dataset is determined by the number of combinations of your system parameters.

Export the dataset to a structured format (e.g., CSV).

In your report: Clearly define the input features and output labels and explain how the dataset was generated.

Task 3.3.2 Machine Learning Model Development in Python **(6 marks)**

In this task, you will develop a simple supervised machine learning model to predict the throughput of an IEEE 802.11 network based on system parameters. Using the dataset generated in Task 3.1, you will train a regression model that learns the relationship between key network parameters and the resulting throughput. Justify your choice of the regression model.

In your report: 1) Describe the structure of your regression model; and 2) Interpret the learned coefficients and explain their physical meaning (e.g., impact of number of stations or $CW_{\min}$ on throughput).